

ReNikud: Audio-Supervised Hebrew Grapheme-to-Phoneme Conversion

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Abstract—Grapheme-to-phoneme (G2P) conversion for Modern Hebrew is needed for applications like text-to-speech (TTS), but is challenging due to the language’s abjad writing system, which leaves vowels largely unwritten, creating widespread ambiguity. Standard approaches first predict vowel diacritics (nikud) to produce International Phonetic Alphabet (IPA) transcriptions, but this is limited: vocalization data is scarce and laborious to produce, it does not specify features such as lexical stress, and it reflects formal grammatical rules rather than everyday spoken pronunciation. Direct sequence-to-sequence IPA prediction, meanwhile, struggles on limited data and produces frequent hallucinations due to a lack of character-level alignment. Our method, ReNikud, overcomes these limitations with two key insights: (1) Weak audio supervision via a phoneme-based automatic speech recognition (ASR) pseudo-labeling pipeline on thousands of hours of unlabeled Hebrew audio, yielding phonemic transcriptions that reflect natural spoken norms without manual annotation. (2) A pseudo-vocalization architecture that predicts IPA phonemes at each character position, enforcing character-level alignment to mitigate hallucinations. Results on existing Hebrew G2P benchmarks and new targeted test suites for spoken Hebrew show that ReNikud surpasses previous state-of-the-art methods. We will release our code and trained models to support further work on Hebrew TTS and speech technologies.

Index Terms—Grapheme-to-Phoneme, Text-to-Speech, Modern Hebrew, Weakly Supervised Learning, Automatic Speech Recognition, Lexical Stress

I. INTRODUCTION

With increasing interest in text-to-speech (TTS) systems for low-resource languages, the modern Hebrew language poses particular challenges. Hebrew is written as an *abjad*—a writing system that normally does not indicate vowel sounds [1]. Consequently, many words are homographs and require semantic context for correct pronunciation. For example, the word מִסְפָּרִים can be read as /mispaʁ'im/¹ (*numbers*), /mispaʁ'ajim/ (*scissors*), or /mesapaʁ'im/ (*telling*). As this ambiguity confounds TTS generations, leading open-source approaches first predict *nikud* (vocalization, i.e., conventional vowel diacritics) to condition synthesis [2], [3].

One issue is the mismatch between written and spoken Hebrew. Firstly, *nikud* does not fully specify phonetic features such as lexical stress, e.g., בֵּירָה may be pronounced as /b'ira/ (*beer*) or /biʁ'al/ (*capital city*). Secondly, *nikud* conventionally reflects traditional grammatical rules that do not match

everyday spoken Hebrew. For instance, the word וּבִדְרֵךְ is formally vocalized and pronounced as /ubad'eʁeʁ/, but in natural speech, it is pronounced /vebad'eʁeʁ/. Similarly, בִּירוּשָׁלַיִם is formally /biʁuʃal'ajim/, but native speakers typically say /bejeʁuʃal'ajim/ [4], [5].

Another issue is a lack of scalable data sources for directly learning G2P. Vocalization models [6], [7] learn from Hebrew text with manually annotated vowel diacritics, data which is scarce and laborious to produce. Similarly, methods such as [3], [8] that learn from IPA annotations are bottlenecked by data availability. Conversely, another abundant source of data on pronunciation exists—unlabelled Hebrew audio—but existing methods can only learn from text.

Finally, while methods using vocalization poorly reflect spoken pronunciation, they have been shown to outperform direct sequence-to-sequence (seq2seq) prediction of IPA [3], the latter of which struggles to learn on limited data and produces frequent hallucinations due to a lack of enforced character-level alignment with the input transcript.

Our method, *ReNikud*², directly addresses these challenges. To reflect spoken norms and unlock abundant training data, we propose a novel pipeline to train G2P with *weak supervision from unlabeled Hebrew audio*. By applying an automatic speech recognition (ASR)-based pseudo-labeling pipeline, we are able to extract pronunciation information and train on thousands of hours of recordings, a scalable approach that effectively learns spoken Hebrew norms. To enforce character-level alignment between Hebrew characters and phonemes, we introduce a *pseudo-vocalization* architecture. Like traditional vocalization, this predicts phonetic content aligned with each character position, but rather than predicting traditional *nikud* symbols it directly predicts character-aligned IPA phonemes. By using the orthographic structure of Hebrew text, this mitigates hallucinations and increases data efficiency relative to seq2seq baselines, as we demonstrate.

Our results on existing Hebrew G2P benchmarks show that ReNikud succeeds in leveraging audio data and better reflecting Hebrew speech. In addition, we propose additional test suites and targeted evaluations to show improvements on difficult phonetic cases reflecting spoken language.

We will release our code, data and trained models to spur work on Hebrew speech technologies.

*Equal contribution

¹We place the IPA stress mark directly before the stressed vowel, following TTS conventions.

²Rethinking *Nikud*

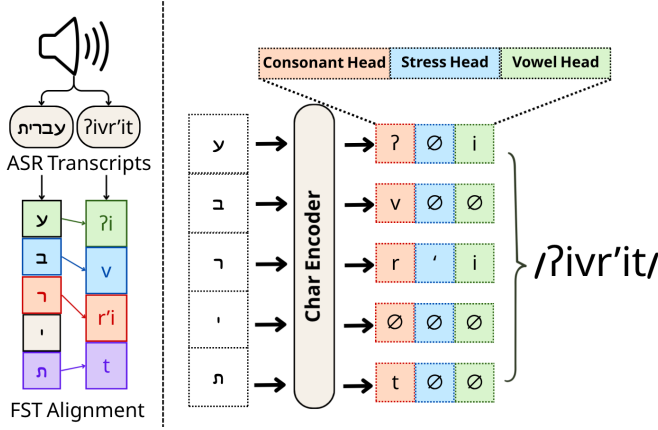


Figure 1: **System overview.** We first pseudo-label audio (left) by creating a many-to-one FST alignment between unvocalized Hebrew text and IPA phonemes derived from two parallel ASR runs applied to Hebrew audio. We then train a pseudo-vocalization architecture (right) where unvocalized Hebrew characters are passed through a character encoder to predict a phonetic triplet (consonant, stress, and vowel) at each position via parallel classification heads.

II. METHOD

The ReNikud pipeline consists of two stages, shown in Figure 1: audio pseudo-labeling (Section II-A), and our pseudo-vocalization architecture trained on this data (Section II-B).

A. Audio Pseudo-Labeling

To learn pronunciation from unlabeled audio at scale, we construct a pipeline that extracts character-aligned IPA annotations using two parallel ASR systems, as shown in Figure 1 (upper left). We extract Hebrew orthographic transcripts with a standard pretrained Hebrew ASR model, and IPA transcripts with a custom ASR model trained to output IPA when applied to Hebrew audio. By applying both ASR systems to a large-scale, unlabeled Hebrew audio corpus, we extract parallel Hebrew text and IPA transcriptions that serve as *pseudo-labels*, providing weak supervision for our downstream G2P model.

Importantly, we retain only pairs where the Hebrew and IPA transcripts agree on the number of words and where every word passes the FST alignment procedure described below. This filtering retains approximately [TODO: XXX%] of the original samples. While strict, this selection criterion still yields 1.52 million aligned sentence from the large dataset, selecting for more reliably transcribed utterances.

Finally, we perform a string alignment process based on the Hebrew orthography’s abjad structure. In general, abjads encode consonants linearly with interleaved, unwritten vowels, meaning that graphemes map monotonically to (consonant, vowel) pairs. As Hebrew also has unwritten lexical stress, each grapheme maps to a *phonetic triplet* encoding such a pair and a binary stress indicator. We find this alignment with a simple finite state transducer (FST) handling known consonant values, including one-to-many (e.g., כ → /b, v/) and many-to-one (e.g.,

ט, ת → /t/) mappings, as well as orthographic complexities such as:

- **Digraphs:** Loanwords in Modern Hebrew frequently use an apostrophe (*geresh*) to denote non-native phonemes (e.g., ץ for /z/; the base letter צ normally represents /z/). The FST assigns the digraph phoneme to the base letter and passes over the *geresh*.
- **Word-final** ח (/x/) may occur with an additional *preceding* /a/ vowel (*patah gnuva*), e.g., לוח /l'uaχ/. We handle this reordering with a special combined /aχ/ value for the vowel slot.
- **Silent letters:** The Hebrew letters ו, י, ה, א may either indicate consonant sounds or may be silent (*matres lectionis*). We handle the latter case with a null (∅) consonant class.

An example of the resulting alignment between Hebrew characters and phonetic triplets is shown in Table I.

Word	Char	Consonant	Vowel	Stress
שלום (/ʃal'om/)	ש	/ʃ/	/a/	0
	ל	/l/	/o/	1
	ו	∅	∅	0
	ם	/m/	∅	0
צִיפּוֹס (/tʃ'ips/)	צ	/tʃ/	/i/	1
	י	∅	∅	0
	פ	∅	∅	0
	ס	/p/	∅	0
	ו	/s/	∅	0
תפוח (/tap'uaχ/)	ת	/t/	/a/	0
	פ	/p/	/u/	1
	ו	∅	∅	0
	ח	∅	/aχ/	0

Table I: **Examples of FST-derived alignment** between Hebrew characters and phonetic triplets (consonant, vowel, stress).

B. Pseudo-Vocalization Architecture

Our goal is to create a Hebrew G2P model that maps unvocalized Hebrew text directly to IPA strings. Because Hebrew is an *abjad*, there is a strong, local relationship between individual written letters and their phonetic realizations. Standard sequence-to-sequence models fail to exploit these localized properties, leading to hallucinations.

To mitigate this, we frame the G2P task as a constrained, per-character classification problem—a method we term *Pseudo-Vocalization*, illustrated in Figure 1 (right). While a single Hebrew character typically corresponds to more than one IPA symbol (e.g., a consonant followed by a vowel), we resolve this by having every Hebrew letter independently predict exactly one phonetic triplet, as described in Section II-A.

The core model is a character-level transformer encoder. On top of this encoder, we attach three parallel, independent classification heads that simultaneously predict the phonetic attributes for each character directly from the encoder’s hidden states:

- **Consonant Head:** Selects from 25 IPA consonants or null ($\emptyset$).
- **Vowel Head:** Selects from 5 vowels, null ($\emptyset$), or the special /ax/ token (see Section II-A).
- **Stress Head:** Binary classifier for lexical stress.

At inference time, realizations are predicted by taking the argmax of logits for each head. In addition, we apply *constrained decoding* to enforce hard constraints on Hebrew letters and phonetic realizations: the argmax is calculated only over possible consonantal realizations of a letter (e.g., $\aleph$ can only be realized as /b/ or /v/). We also enforce a word-level constraint that exactly one lexical stress is predicted.

C. Renikud Classifier

The core model is built upon dicta-il/dictabert-large-char, a character-level BERT encoder pre-trained on Hebrew. We selected this base encoder because our method requires character-level representation. On top of this base encoder, we attach three independent classification heads that simultaneously predict the following phonetic elements for every input character:

- **Letter Head:** Predicts the base consonant from a predefined set of 25 possible classes.
- **Vowel Head:** Predicts the associated vowel from a set of 7 possible classes.
- **Stress Head:** A binary classifier indicating whether the character carries primary lexical stress (true/false).

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

For our base ASR model, we employ the Whisper Large v3 Turbo checkpoint adapted for Hebrew by ivrit.ai, which we subsequently fine-tune on text-and-IPA objective. Our training process is divided into two distinct stages:

Dataset	Size	Role
SASpeech [9]	~18h	Pretraining (Stage 1)
Recital	~46h	Pretraining (Stage 1)
ILSpeech [3]	~2h	Fine-tuning (Stage 2)

Table II: **Data sources** for IPA ASR training.

Stage 1: Pretraining. During the initial pre-training phase, we utilized two primary corpora. The first is a subset of the *SASpeech* dataset [9]. While the full corpus contains roughly 30 hours of audio, the automatic audio files that contain 26 hours we filtered short files with single word, we used the ivrit’s ASR to filter all the audio with a lot of “umm..” by doing this we trim it down to approximately 14 hours of the automatically aligned data. with the 4 manual data we got about 18 hours. The second is the crowd-sourced *Recital* dataset, consisting of approximately 46 hours of raw audio. we processed the datasets’ existing Hebrew transcripts directly through Phonikud [3] to produce IPA.

Stage 2: Fine-tuning and Evaluation of the ASR Following pre-training, the model was fine-tuned on the manual

ipa *ILSpeech* dataset [3], which comprises approximately 2 hours of audio. From the dataset, we extracted exactly 150 recordings for our held-out test set, reserving the remainder for fine-tuning.

To assess the ASR-IPA capabilities, we evaluated the fine-tuned model on two datasets: the held-out test set of 150 manually annotated sentences from *ILSpeech*, and *Michel*, a synthetic out-of-domain dataset comprising 250 sentences. We report Character Error Rate (CER), Word Error Rate (WER), Vowel Error Rate (VER), and Stress Error Rate (SER). Results are detailed in Table III.

Test Set	CER	WER	VER	SER
ILSpeech (150 seq.)	0.0240	0.1120	0.0224	0.9240
Michel (250 seq.)	0.0478	0.2216	0.0321	0.1767

Table III: **ASR IPA evaluation results** on the in-domain ILSpeech and out-of-domain synthetic Michel datasets.

[Max: is 350 sentences enough for showing the results?]
[Max: is there a way to explain that michel is a legit dataset?]

Stage 3: Training of Renikud For our primary large-scale dataset, we utilized Knesset Vox [10], specifically drawing upon approximately 1.7k hours of parliamentary recordings from its training set. Knesset Vox has ready to use transcripts by using ASR and forced alignment between the already made scripts from the kneset parliament transcripts. Because the Whisper ASR model experiences performance degradation on longer audio inputs (exceeding 25 seconds), we bypassed the original Knesset Vox sentence annotations and re-segmented the raw audio into optimized 5–15-second clips to ensure correct results of phonetic transcriptions with the ASR. We then processed these shorter segments using both a standard Hebrew Whisper model and our custom IPA-tuned Whisper model to extract parallel Hebrew text and IPA transcriptions. After extracting the pseudo-labels and applying the FST alignment process described in Section II-A, To ensure memory stability and prevent out-of-memory (OOM) errors during training, we filtered out sentence-length outliers using the standard $1.5 \times \text{IQR}$ rule. Specifically, we discarded sequences with lengths falling outside the bounds of $[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]$. Ultimately, this weak supervision pipeline yielded a dataset of approximately 1.52 million aligned Hebrew-to-IPA sentences. aligned Hebrew-to-IPA sentences. [Morris: does Knesset Vox have gold transcripts? do we ignore them?] [Max: Yes, but we create shorter clips of 5-15 and it will be a headache to use the gold labels also probably were made with whisper] [Morris: should be explained concisely][Max: added some explanation about the transcriptions]

Stage 4: Evaluation of Renikud To benchmark our proposed ReNikud architecture, we evaluated its grapheme-to-phoneme mapping capabilities against the standard Phonikud baseline.

B. MILIM Testset

MILIM³ testset consists of a targeted evaluation corpus de-

³Benchmark for Individual-word Testing of Underspecified IPA

Cat.	Input	Target	IPA	Cat.	Input	Target	IPA
Gender	תגיד, הכל בסדר איתך?	איתך	/ʔit'aχ/	Names	ליה איפה החולצה הכחולה שלי?	ליהי	/li'hi/
Acronyms	הוא שירת באמ"ן	באמ"ן	/beʔam'an/	Homogr.	הוא קרא ספר מעניין בזמן שהוא ישב אצל ספר נשים.	ספר	/s'efek/-/sap'ak/
Penult.	קניתי לחם טרי	לחם	/l'eχem/	Min. Str.	/ היה לו נחת רוח מההישגים של ילדיו המטוס נחת בשלום על המסלול	נחת	/n'aχat/-/naχ'at/
Rare Ph. Foreign	הבאתי ציפס בשבילך, תאכל מהר. איזה וויב טוב יש פה	ציפס וויב	/tʃ'ips/ /v'ajb/	Slang	הוא בן אדם גנוב לגמרי.	גנוב	/gan'uv/

Table IV: Selected examples from MILIM, split by category (Cat.). Abbreviated category names are penultimate stress (Penult.), rare phonemes (Rare Ph.), homographs (Homogr.), and minimal stress pairs (Min. Str.).

signed to systematically assess a model’s capacity for complex linguistic disambiguation in context-dependent orthographies. The testset has 1,653 sentences about 150 for each category some categories like ilspeech and homographs contain multiple words so the testset is on 3,110 changing words. MILIM testset except of the ilspeech, the generation of the sentences was created using Gemini-3.1-pro and then we fixed it manually. the high result of the gemini in the graphs might be the results of bias towards his own sentences. ILspeech was updated to version 2 to fix mistakes like e.g. אוואטרים, ותוכן, it was אבטרים.

Targeted Evaluation Methodology:

The dataset is categorized into distinct linguistic challenges:

- **Morphological and Syntactic Ambiguity (gender):** Hebrew pronouns, verbs, and adjectives frequently share identical orthography but shift in pronunciation based on the gender of the subject or object.
 - Example: אני נותן לך את הספר תקרא אותו (לך) → /lex'a/ (Resolved via the masculine future verb תקרא).
- **Semantic Ambiguity (homographs):** This category tests identically spelled words with completely divergent meanings and phonetic structures.
 - Example 1: חתכתי בצל למרק (I cut an onion for the soup) → /bats'al/.
 - Example 2: ישבנו בצל העץ (We sat in the shade of the tree) → /bets'el/.
- **Lexical Slang (slang):** Modern spoken Hebrew frequently diverges from standard dictionary vocabulary, adopting informal, non-standard words.
 - Example: איזו פאדיחה הייתה אתמול (fad'ixa)
 - Targeted Label: פאדיחה → /fad'ixa/
- **Colloquial Phonology (colloquial):** Everyday spoken Hebrew often alters normative vowel patterns or consonant articulation, reflecting register-specific shifts rather than lexical slang.
 - Example: הוא קנה אופנוע חדש ונוסע איתו לעבודה (Pre-scriptive Formal: /ʔofan'oa/)
 - Targeted Label: Index 3 (אופנוע) → /ʔofan'oa/ (Pre-scriptive Formal: /ʔofn'oa/).
- **Foreign Loanwords and Rare Phonemes (loanwords):** Hebrew orthography adapts to non-native phonemes (e.g., /w/, /d/, /t/) using modifying characters (geresh) or unusual letter combinations.

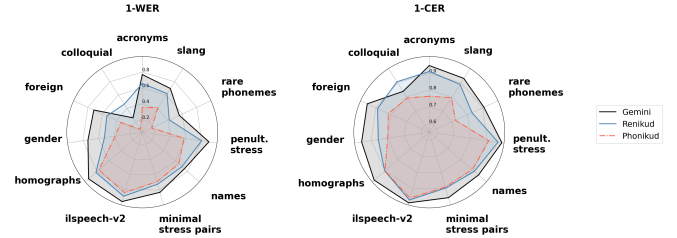


Figure 2: Word accuracy rate (left) and character accuracy rate (right). [Max: should i add the name Gemini-3.1-pro? in the graph or in caption?] [Morris: in caption or paper text is fine. It should be a gray dotted line, not solid black, and in caption we should say that it provides an upper bound]

- Example: הזמנתי המבורגר וציפס
- Targeted Label: ציפס → /tʃ'ips/

- **Baseline Control (ilspeech-v2-test):** Main testset of phonikud and also to test general speech classification correction.

- Example: הייתה מעורבת ממילא באינספור פרויקטים צבאיים מתקדמים [Max: i will add here the ipa but i dont think we need it /hajt'a meʔox'evet mim'ele beʔejnsf'ox psoj'ektim tsvaʔij'im mitkadm'im/]

Evaluation Methodology: Predictions were compared against human-annotated gold-standard IPA transcriptions. Performance is reported via Word Error Rate (WER) and Character Error Rate (CER). The overall model performance (reported as OVERALL) is calculated as a micro-average of the error rates across all distinct evaluation categories, encompassing both the challenge suite and the generalized test set in [3] as seen in

Results and Analysis: The comparative results are detailed in Table VII. For example, ReNikud accurately predicts the colloquial slang פאדיחה as /fad'ixa/, whereas the Phonikud incorrectly outputs the /padiχ'a/. Furthermore, ReNikud successfully maps non-native phonemes in foreign loanwords, correctly resolving the English 'w' in ווינר (/w'ineχ/) and the 'j' [Max: literally the j not yod] sound in ג'וב (/hadz'ob/), which the Phonikud misclassifies as (/vin'ek/ and /hadz'ov/). Gemini-3.1-pro results used as an upper bound for the possible results. In the colloquial category, Gemini significantly underperformed compared to ReNikud. This discrepancy arises because Gemini’s phonetic knowledge is largely derived from

prescriptive, text-based dictionaries and formal grammatical rules, whereas ReNikudlearns directly from the acoustic distributions of authentic spoken Hebrew. A prominent illustration of this is the handling of the Hebrew conjunction ׀ (“and”). According to strict normative grammar, the conjunction shifts to the /u/ phoneme. However, modern spoken Hebrew routinely ignores this rule, favoring the default /ve/ pronunciation. For example, given the sentence הכל תלוי בחברים ומשפחה תומכת (Index 3: ומשפחה), Gemini erroneously outputs the prescriptive /umiʃpax'a/. In contrast, ReNikudcorrectly predicts the targeted colloquial pronunciation /vemiʃpax'a/, demonstrating its superior alignment with real-world native speech patterns.

C. Vocalization (Nikud Prediction)

G2P is similar to diacritization which is an important task, but nikud data is not scalable unlike audio. We want to demonstrate with our per-character architecture finetuning on nikud is not only possible its also better than using our method as shown in 3. We evaluated its capacity to perform standard Hebrew text diacritization (*nikud*). While our primary G2P pipeline scales by leveraging audio supervision, diacritization remains a foundational task in Hebrew natural language processing (NLP) that shares the exact same core structural challenge: resolving highly localized, context-dependent ambiguities at the grapheme level. Following the per-character classification methodology established by DictaBERT [7], [Morris: need better explanation] [Max: is this good enough?] we retained our pre-trained ReNikud encoder but replaced the phonetic classification heads with a head tailored to directly predict the correct *nikud* class.

This adapted model was fine-tuned on a dataset of 5 million unvocalized sentences taken from the Knesset corpus that was used in Phonikud training data (Phonikud’s data was generated from Dicta Nakdan API) but without the stress, shva and the prefix tags. From the 5m sentences we sampled 10k, 25k, 50k and 100k sentences and other 10k sentence for validation. For evaluation we took 100 sentences from nakdimon test and manually fixed the 100 sentences. **Test Set Size and Model Convergence:** To verify that 100 sentences provide enough data to accurately evaluate the models, we used paired Wilcoxon tests and Bootstrap resampling to measure the statistical differences at each stage.

Test Set Size and Model Convergence: To verify that 100 sentences provide enough data to accurately evaluate the models, we used paired Wilcoxon tests and Bootstrap resampling to measure the statistical differences at each stage.

- **10k Steps (Detecting Differences):** Early in training, the test set easily detected a clear performance gap. ReNikud outperformed Dicta with a Word Error Rate (WER) of 13.4% compared to 15.7%. The statistical test confirmed this was a real difference ($p < 0.001$).
- **25k and 50k Steps (Convergence):** As the models trained on more data, their performance leveled out. As early as 25k steps, the statistical difference between the models was eliminated ($p = 0.669$). By 50k steps, their

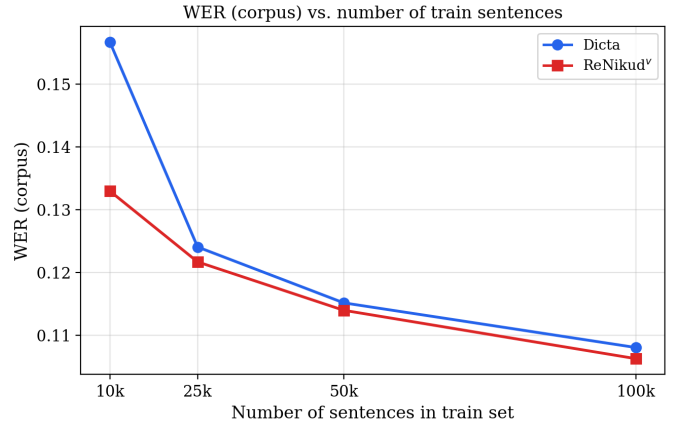


Figure 3: Diacritization performance on test set over the different number of sentences in the training

WER scores were nearly identical (11.5% vs. 11.6%, $p = 0.844$), confirming the models were converging.

- **100k Steps (Statistical Parity):** At the final 100,000-step mark, both models reached a complete statistical tie (WER of 10.7% vs. 10.9%). More importantly, the Bootstrap Confidence Interval narrowed to $[-0.013, 0.011]$. This tightly bound interval means we are 95% sure that the maximum possible difference between the two models is around $\pm 1.3\%$.

[Max: how do we want to show the qualitative results? the models results are almost 1 to 1.]

We trained the subsets on ReNikudcheckpoint and on dicta-il/dictabert-large-char checkpoint the training params as follows: same hyperparams with 5 early-stopping every epoch until converges. and compared the results of the test as seen in

The comparative results against the DictaBERT-Menaked⁴, Dicta’s best model for nikud show that using ReNikuda bit better convergence with small amount of data and the final result are close to dicta’s best public model as shown in VI.

[Max: do we even want to compare it to DictaBERT-Menaked?]

D. Ablation and Architectural Comparison

To isolate the contributions of our core design choices and validate the efficacy of our approach, we designed evaluations to justify two key components: our novel classification architecture and the necessity of audio-derived weak supervision. The comparative results of these evaluations are detailed in Table VIII.

Architectural Ablation: To demonstrate that our independent classification head is superior for Hebrew G2P, we compared it against two standard architectures trained on the identical weakly-supervised Knesset Vox dataset. The baselines included a sequence-to-sequence approach (ByT5-Small) and a Connectionist Temporal Classification (CTC)

⁴dicta-il/dictabert-large-char-menaked

	Gender	Acronyms	Penult. Stress	Rare Phonemes	Foreign	Names	Stress Homog.	Slang	Colloquial
ReNikud (Ours)	47.4/ 14.3	34.2 /8.9	19.9 /3.6	58.3 /17.0	46.5/11.1	29.3 /9.5	18.2 /9.8	37.8 /11.2	53.6 /9.3
CTC Loss Network	62.5/20.7	34.9/ 8.7	21.2/3.8	63.6/17.2	44.5 /10.6	31.3/10.4	22.7/10.8	38.5/11.6	57.6/10.1
ReNikud (Phonikud Data)	45.4 /15.0	76.3/25.9	29.8/6.3	82.1/29.1	47.1/12.6	32.7/11.0	21.2/10.5	43.6/15.1	90.1/20.8
Renikud w/Vox labeled with Phonikud	64.5/20.7	80.3/26.3	33.1/7.0	84.1/27.0	57.4/15.9	34.0/11.1	23.7/10.9	55.1/19.7	92.1/20.7
Seq2Seq (ByT5)	61.2/24.0	43.4/10.9	27.8/6.7	69.5/20.1	55.5/15.8	36.7/13.4	29.1/15.7	51.9/20.3	55.6/10.3
Phonikud (Baseline)	61.2/20.0	64.5/23.6	42.4/9.1	82.8/27.9	65.8/18.2	34.7/11.1	23.2/10.0	59.0/20.6	91.4/20.8

Table V: Detailed evaluation on the Hebrew G2P challenge suite (WER/CER, in %). Our proposed ReNikud model trained on the Knesset Vox corpus achieves the best overall performance among dedicated G2P models, significantly reducing errors in complex domains such as Slang and Rare Phonemes compared to both the Phonikud baseline and a variant of ReNikud restricted to the same legacy training data as Phonikud. Best results per column are highlighted in bold.

Model	Train Size	WER (Genus)	CER (Genus)	WER (Phonikud)	CER (Phonikud)
DictaBERT-Menaked	(pretrained)	1.68%	29%	1.84%	29%
ReNikud^V (Ours)	100k	1.06%	1.84%	1.84%	1.84%
Dicta	100k	1.06%	1.84%	1.84%	1.84%

Table VI: Diacritization performance on the test set

network built on the same dicta-il/dictabert-large-char base encoder used in our method.

Qualitative Analysis of Autoregressive Hallucinations:

Qualitative analysis of the predictions highlights a critical flaw in the Seq2Seq baseline: autoregressive hallucination. Unlike classification-based models that map predictions strictly to the input characters, Seq2Seq models frequently generate phonetic sequences that completely mismatch the provided Hebrew orthography. Our constrained, 1-to-1 per-character mapping architecture structurally prevents these failures by forcing the sequence length and alignment to match the source text.

Qualitative Analysis of Phonetic Misalignment:

Qualitative analysis reveals a critical limitation in the Seq2Seq baseline: phonetic and prosodic misalignment. Unlike classification-based models that strictly map predictions to input characters, unconstrained Seq2Seq generation frequently breaks the structural boundaries of the target words. Rather than producing full lexical hallucinations, the model suffers from severe localized distortions—specifically, inserting extraneous vowels, omitting required vowels, and incorrectly placing lexical stress.

An analysis of the slang and colloquial subsets exposes the severity of these generative errors. For instance, ByT5 frequently **inserts extraneous phonemes** to force unfamiliar words into normative grammatical templates. When predicting the slang term לאפה (/l'afa/), the model hallucinates **[Max: how would you call it?]** a complex glottal structure, outputting /leʔap'a/. Similarly, for the word מאמי (/m'ami/), it erroneously generates /maʔam'i/.

Conversely, the model frequently **omits required vowels**, artificially compressing words. For the word בלאגן (/balag'an/), the model completely drops the initial vowel, generating

/bl'agan/ (באגן) instead of /balag'an/ (בלאגן), skipping the vocalization entirely, outputting /bɪʔan/. Finally, the autoregressive decoding mechanism repeatedly loses track of the word's prosodic structure, **misplacing the primary stress marker** (the /' / symbol) onto the wrong syllable. This is particularly prevalent in penultimate-stressed slang, where ByT5 defaults to ultimate stress—predicting /lig'a/ instead of /l'iga/ (ליגה) and /temb'el/ instead of /t'embel/ (טמבל).

Our ReNikud architecture structurally prevents these failures. By enforcing a constrained, 1-to-1 per-character mapping, the classifier guarantees strict structural alignment. This ensures that every orthographic character is explicitly accounted for with a discrete phonetic label, inherently preventing the runaway generation of extra vowels, the dropping of required phonetic slots, and anchoring the stress marker directly to the source text. **Qualitative Analysis of Model Predictions:**

Qualitative analysis of the test suite demonstrates ReNikud's ability to successfully resolve complex linguistic challenges where both alignment-free architectures (ReNikud-CTC) and legacy classification baselines (Phonikud) fail:

- **Morphological and Syntactic Ambiguity (gender):** The baseline models fail to use the context of the feminine verbs and pronouns in the sentence to correctly predict the feminine suffix /eχ/.
 - *Example:* המפתח נשאר אצלך את בטוחה (The key stayed with you [f], are you [f] sure?)
 - *Targeted Label:* אצלך → /ʔetsl'eχ/
 - *ReNikud (Ours):* /ʔetsl'eχ/ (Correctly resolves feminine context)
 - *Baseline Failures:* ReNikud-CTC predicts /ʔetslɔχ/ (mispredicts the masculine suffix and entirely drops the stress marker); Phonikud predicts /ʔ'etslɔχ/ (invalid phonetics).
- **Semantic Ambiguity (homographs):** The baselines frequently fail to resolve identically spelled words based on sentence-level context, often defaulting to the most common statistical pronunciation.
 - *Example:* הוא ישב לנוח מעט בצל העץ הגדול ואכל לחם (He sat to rest a bit in the **shade** of the עם בצל) (He sat to rest a bit in the **shade** of the

large tree and ate bread with **onion**.)

- *Targeted Labels*: בצל → /bets'el/ (shade); בצל → /bats'al/ (onion).
 - *ReNikud (Ours)*: /bets'el/, /bats'al/ (Perfectly distinguishes both contexts).
 - *Baseline Failures*: ReNikud-CTC incorrectly predicts /bets'el/ for both words, missing the contextual shift. Phonikud suffers from vowel reduction and misplaces the stress on the second word (/bts'el b'atsal/).
- **Colloquial Phonology (colloquial)**: Legacy models heavily bias toward prescriptive grammar rules, failing to capture how native speakers actually pronounce conjunctions and prepositions.
 - *Example*: היו רעמים וברקים חזקים בלילה (There were strong thunders and **lightnings** at night.)
 - *Targeted Label*: וברקים → /vebʁak'im/ (Colloquial pronunciation).
 - *ReNikud (Ours)*: /vebʁak'im/
 - *Baseline Failures*: Both ReNikud-CTC and Phonikud erroneously apply the formal "bumaf" rule, shifting the conjunction to the normative /u/ (/ubʁak'im/ and /uʁak'im/), which is highly unnatural in spoken modern Hebrew.
 - **Lexical Slang (slang)**: Baselines struggle to assign correct vowel templates to non-standard vocabulary, frequently mispredicting the vowels.
 - *Example*: מה קורה **כפרה** עליך (What's happening, **darling** [slang].)
 - *Targeted Label*: כפרה → /kap'aʁa/
 - *ReNikud (Ours)*: /kap'aʁa/
 - *Baseline Failures*: ReNikud-CTC predicts /kapeʁ'a/ (hallucinating an 'e' vowel and shifting stress); Phonikud predicts /kapaʁ'a/ (shifting stress to the final syllable).
 - **Acronyms (acronyms)**: The baseline models struggle with correct stress placement and vowel resolution for military or medical acronyms.
 - *Example*: הוא מע"ר במד"א (He is a medic in MADA)
 - *Targeted Label*: במד"א → /bem'ada/
 - *ReNikud (Ours)*: /bem'ada/
 - *Baseline Failures*: ReNikud-CTC suffers from prosodic duplication, predicting /bem'ad'a/ with two conflicting stress markers; Phonikud predicts /be-mad'a/, defaulting to standard ultimate stress.
 - **Penultimate Stress (loanwords)**: Legacy baselines heavily bias towards final ultimate stress (Mil'ra), frequently failing on modern loanwords that require penultimate stress (Mil'el).
 - *Example*: הרעיון מבוסס על **קונספט** שונה (The idea is based on a different **concept**.)
 - *Targeted Label*: קונספט → /k'onsept/
 - *ReNikud (Ours)*: /k'onsept/
 - *Baseline Failures*: Both ReNikud-CTC and Phonikud predict /kons'ept/, incorrectly forcing

the foreign loanword into a native ultimate-stress pattern.

- **Rare Phonemes (rare phonemes)**: Baseline models struggle to map orthographically modified characters (like the Geresh) to their correct phonetic outputs.
 - *Example*: זה לא הז'אנר שלך אתה לא תאהב (It's not your **genre**, you won't like it.)
 - *Targeted Label*: הז'אנר → /haʒ'anɛʁ/
 - *ReNikud (Ours)*: /haʒ'anɛʁ/
 - *Baseline Failures*: ReNikud-CTC mispredicts the internal vowel cluster, outputting /haʒ'enɛʁ/; Phonikud misplaces the prosodic stress, predicting /haʒan'eʁ/.
- **Phonetic Insertion and Misalignment (foreign)**: Because the CTC baseline must dynamically manage blank tokens to achieve sequence alignment, it frequently suffers from phonetic distortion, inserting nonexistent syllables.
 - *Example*: איזה האקר פרץ לאתר (Some **hacker** broke into the site.)
 - *Targeted Label*: האקר → /h'akeʁ/
 - *ReNikud (Ours)*: /h'akeʁ/
 - *Baseline Failures*: ReNikud-CTC predicts /haʔak'eʁ/, hallucinating an inserted glottal stop (/ʔ/), adding an extra syllable, and shifting the stress. Phonikud predicts /hak'eʁ/ (incorrect ultimate stress).

Audio Supervision vs. Text Data: To justify our weak supervision pipeline, we compared the ReNikud architecture trained on Knesset Vox that the ipas were created by phonikud (Renikud w/Vox labeled with Phonikud) and text-derived Phonikud dataset (ReNikud (Phonikud Data)) as shown in Table VII and ?? the results are worst in any category then using Knesset Vox with audio labels.

Training Details: All models were continuously evaluated using a validation set of 250 gold sentences. [Max: what is expected from this part?]

Ablation Results: The results in Table VIII confirm the independent contributions of both our direct phonetic classification architecture and our weak-supervision data pipeline. Structurally, when compering different data on the same model we can see that using Phonikud's data or using the same knesset vox data but not using ASR-IPA for IPA labels but using phonikud for labeling resulting with worst WER and CER overall results and across all the categories in MILIM as shown in Table VII

by taking seq2seq and a network with ctc loss the results are the same except of 2 categories that the ctc loss is slightly better but when overall results are taken to account ReNikudis better.

IV. RELATED WORK

Explicit G2P conversion for Hebrew was recently introduced by Kolani et al. [3], following prior works on Hebrew diacritization [6], [7]. These approaches are bottlenecked by the availability of annotated textual data, while we use audio as a scalable source of pronunciation information.

	Gender	Acronyms	Penult. Stress	Rare Phonemes	Foreign	Names	Stress Homog.	Slang	Colloquial
ReNikud (Ours)	47.4/ 14.3	34.2 /8.9	19.9 /3.6	58.3 /17.0	46.5/11.1	29.3 /9.5	18.2 /9.8	37.8 /11.2	53.6 /9.3
CTC Loss Network	62.5/20.7	34.9/ 8.7	21.2/3.8	63.6/17.2	44.5 /10.6	31.3/10.4	22.7/10.8	38.5/11.6	57.6/10.1
ReNikud (Phonikud Data)	45.4 /15.0	76.3/25.9	29.8/6.3	82.1/29.1	47.1/12.6	32.7/11.0	21.2/10.5	43.6/15.1	90.1/20.8
Renikud w/Vox labeled with Phonikud	64.5/20.7	80.3/26.3	33.1/7.0	84.1/27.0	57.4/15.9	34.0/11.1	23.7/10.9	55.1/19.7	92.1/20.7
Seq2Seq (ByT5)	61.2/24.0	43.4/10.9	27.8/6.7	69.5/20.1	55.5/15.8	36.7/13.4	29.1/15.7	51.9/20.3	55.6/10.3
Phonikud (Baseline)	61.2/20.0	64.5/23.6	42.4/9.1	82.8/27.9	65.8/18.2	34.7/11.1	23.2/10.0	59.0/20.6	91.4/20.8

Table VII: Detailed evaluation on the Hebrew G2P challenge suite (WER/CER, in %). Our proposed ReNikud model trained on the Knesset Vox corpus achieves the best overall performance among dedicated G2P models, significantly reducing errors in complex domains such as Slang and Rare Phonemes compared to both the Phonikud baseline and a variant of ReNikud restricted to the same legacy training data as Phonikud. Best results per column are highlighted in bold.

Method	Validation Set (250)		Test Set (1,500)	
	WER	CER	WER	CER
Seq2Seq	24.1	4.9	32.1	11.0
ReNikud (Phonikud Data)	23.4	6.8	32.6	11.0
ReNikud (w/Vox labeled with Phonikud)	20.7	4.5	33.0	9.9
CTC Loss Network	21.2	3.8	27.9	8.9
New Classifier Head (Ours)	13.7	2.7	26.7	8.5

Table VIII: Performance comparison of generation methods. Error rates are reported as percentages for Word Error Rate (WER) and Character Error Rate (CER), evaluated on both a curated validation set of 250 sentences and the full held-out test set of 1,500 sentences.

Among prior works learning pronunciation from audio in other language settings, we distinguish between two approaches:

(1) **Audio-supervised** methods such as ours use audio during training to improve pronunciation knowledge. Most similar to our work, a few studies use audio to improve G2P for English [11], [12] or in multilingual settings [13]. However, these works train on labeled audio as a supplementary signal to labelled text, while we use large-scale unlabeled audio as our primary training signal. Additionally, we operate on the Hebrew language which has a high degree of orthographic ambiguity and phonetic features unspecified in text (stress, spoken norms, etc.), and our pseudo-vocalization architecture avoids seq2seq hallucinations by enforcing abjad-style alignment.

(2) **Audio-guided** methods also take audio along with text at inference time for G2P [14]–[16] or abjad diacritization [17], [18]. While adding audio as an additional input can provide a richer signal, we focus on the case where only text is available at inference time.

V. CONCLUSION

We introduced ReNikud, a *Pseudo-Vocalization* architecture for Hebrew G2P conversion. By reframing the generative sequence-to-sequence problem into a per-character phonetic triplet classification task (Consonant, Vowel, Stress), our method structurally mitigates the generative hallucinations observed in standard Seq2Seq models. [Morris: this paragraph

should be changed, renikud is also a method for using audio, not just the architecture] [Morris: change claim to be about reducing degrees of freedom]

[Yakov: Add a mention about the limitation where, due to our strict alignment, the aligner might drop some informal variants such as , which might be pronounced as jixtov.]

To train this architecture, we developed a weak-supervision pipeline utilizing a monotonic FST to align continuous IPA transcriptions from an adapted Whisper ASR model with unvocalized Hebrew graphemes. Our evaluations demonstrate that ReNikud lowers Word Error Rates compared to the legacy Phonikud baseline and the evaluated Seq2Seq and CTC architectures [Morris: phrase that part better], particularly on complex edge-cases such as colloquial slang, rare phonemes, and penultimate stress. Furthermore, we validated the extensibility of our core character-level encoder by successfully adapting it for Hebrew text diacritization via knowledge distillation, using only a fraction of standard training data volumes. [Morris: llm speak]

Limitations and Future Work: A primary limitation of our methodology is its reliance on ASR-generated pseudo-labels, as inherent transcription errors from the ASR model naturally propagate into the G2P training data. Additionally, because the Knesset Vox dataset consists exclusively of parliamentary speeches, the training corpus carries a distinct formal and demographic bias. [Morris: LLM speak] This restricts the model’s exposure to diverse, casual, and second-person conversational phonology—a domain shift that accounts for the observed performance gap in resolving gendered morphology compared to legacy models trained on broader literary text. [Morris: are you sure that’s true?] Future work will focus on expanding our weak-supervision pipeline to incorporate more diverse, unstructured conversational audio corpora to address these domain-specific biases. Furthermore, because the core challenge of unwritten vowels is shared across other abjad writing systems (such as Arabic), we plan to extend the *Pseudo-Vocalization* architecture to these languages to evaluate its cross-lingual generalizability. [Max: something do to with חֵם וְנוֹבוֹת] [Morris: what do they limit?]

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